aquamarined07@gmail.com · github.com/Patcholie · patcholie.github.io · linkedin.com/in/aviv-shaked · Haifa, Israel · 054-543-5388

CYBERSECURITY DEVELOPER & RESEARCHER · HAIFA, ISRAEL

AVIV SHAKED

C++20 systems programmer and cyber researcher with deep expertise in network protocols, covert channels, and binary exploitation. Graduate of **Magshimim+** — Israel's elite advanced cybersecurity track. Architect of **HitchHiker**: a production steganography platform with zero observable network presence. **2nd globally**, AI4Y 2023.

C++20 Systems · Network Protocols · Binary Exploitation · Reverse Engineering · Magshimim+ · ◆ AI4Y World 2023 · 2nd

SKILLS

LANGUAGES

C++20 · Python · x86-64 ASM · C# · SQL

C++ & SYSTEMS DESIGN

RAII / Memory Safety · OOP Architecture · Interface Abstraction · CMake / Build Systems · Multi-threading · Modern C++20 Idioms

NETWORKING

TCP/UDP/IPv4 · Socket Programming · DNS / ICMP / ARP / DHCP · HTTP/HTTPS/TLS · iptables / NFQueue · Wireshark

SECURITY

Binary Exploitation · Reverse Engineering · Net Steganography · ROP Chains · Protocol Attacks · Threat Modeling · HackTheBox · Web Security

CRYPTOGRAPHY (APPLIED)

ECDH / ChaCha20 / AES · HKDF / Argon2id / KDFs · Hash Exploits · Crypto Attacks

TOOLS

Binary Ninja	IDA Pro
GDB / objdump	Wireshark
Docker	CMake
Git	Linux / Bash
OpenVPN	AWS EC2
JetBrains	nmap

EDUCATION

Magshimim Cyber

Advanced Track · Magshimim+

Israel's elite national cybersecurity program · 2023–2026

BOSMAT, Haifa

CS + AI · 10 study units (max) · 2020–2026

PROJECTS

HitchHiker

C++20 · 2025–2026 · Completed

Network Steganography Platform — co-developed with Daniel Varkin

C++20 · Crypto++ · libnetfilter_queue · FTXUI · Redis · SQLite · Docker · OpenVPN · AWS EC2 · CMake

- Architected a covert communication platform embedding encrypted messages into live traffic — **zero new packets generated**, no observable network presence
- Full crypto pipeline: ECDH → ChaCha20 → HKDF → Argon2id; bit-precision embedding across IPv4, TCP, UDP, DNS headers via kernel-level `libnetfilter_queue` + iptables
- **Pluggable protocol carrier architecture** — three independent CMake targets (`lib / server / client`) behind a unified `HitchHikerAPI`; deployed on AWS EC2 via Docker

Multiplayer Trivia Game System

C++ / C# / Unity · Magshimim · 2024

C++ · C# · Unity · SQLite · TCP Sockets

- Designed a custom application-layer protocol from scratch; concurrent TCP server supporting **25+ simultaneous players**
- Integrated C++ backend, C# Unity GUI client, and SQLite database — full-stack with clean OOP architecture throughout

Analisis

Python · 2023 · Public (AGPL v3.0)

AI Wound Treatment Platform ◆ 2nd — AI4Y World 2023

Python · OpenAI API · AI/ML

- Co-led a team to build an AI-powered wound diagnosis platform — concept through working prototype
- **Placed 2nd globally** in the AI4Y World Competition (2023) against international competitors

EXPERIENCE

Tools Developer & Product Uploader

2022–2023 · Remote

AllBikes

- Built Python automation tools for web scraping supplier data, cleaning, and structuring product catalogs — **tooling adopted by the full team**
- Operated independently in a professional remote environment throughout secondary school

HONORS & AWARDS

- **2nd Place — AI4Y World** — 2023 · Global
- **Magshimim+ Advanced Track** — 2026
- Magshimim Cyber Graduation — 2026
- Topstep Trading Certificate — 2025–2026

ADDITIONAL

- Active investor & day trader (2+ years) — fundamental & technical analysis, orderflow trading · **Topstep Trading Certificate holder**